

Multiuser Access via OFDMA Technology in High Density IEEE 802.11ax WLANs: A Survey

Saloua BRAHMI, Mohand YAZID and Mawloud OMAR

Research Unit LaMOS (Modeling and Optimization of Systems), Faculty of Exact Sciences,
University of Bejaia, 06000 Bejaia, Algeria

Emails: brahmisaloua1@gmail.com, yazid.mohand@gmail.com, mawloud.omar@gmail.com

Abstract—IEEE 802.11ax known as HEW (High Efficiency WLANs: Wireless Local Area Networks) is the new standard designed to meet the objectives of the next generation of high density wireless networks, such as: airports, bus stations, stadiums, etc. The key innovative feature introduced in IEEE 802.11ax is the OFDMA (Orthogonal Frequency Division Multiple Access) technology. The basic principle of OFDMA at PHY (physical) layer is dividing a radio channel into smaller sub-channels, and hence enabling multiple and independent transmissions between multiple wireless devices. At MAC (Medium Access Control) layer, the OFDMA technology is thus intended to allow a massive connection of users, while reducing access delay and increasing individual throughput. This is why, several OFDMA MAC protocols, based on different approaches, have been designed and implemented for the purpose of efficiently using the radio resources supplied by the OFDMA technology at PHY layer. In this paper, we aim at: (i) reviewing the main existing OFDMA MAC protocols available in the literature with the goal of highlighting the strengths and weaknesses of each protocol, (ii) proposing a new classification of OFDMA MAC protocols, and (iii) drawing some perspectives for future research.

Index Terms—IEEE 802.11ax Standard, High Density WLANs, OFDMA Technology, MAC Protocols, Classification.

I. INTRODUCTION

Wireless local area networks (WLANs), developed over the years from 1999, are being used for communication, exchange of information, business, shopping and several others domain. This is why, the wireless networks are now considered as a technological advance, particularly in the context of the information society, which is one method of overcoming the problem of geographical dislocation [1].

The WLANs are covered by the IEEE 802.11 standard that defines the two lower layers of the OSI (Open Systems Interconnection) model, namely: the PHY (Physical) layer and the MAC (Medium Access Control) layer [2]. The IEEE 802.11 standard have abide many transformations and changes of operating mode to meet the requirements of users, especially in terms of throughput and access delay [3]. Among the IEEE 802.11 amendments [4], we can distinguish: 802.11b proposed in 1999, it provides a peak physical rate of 11 Mbps using the DSSS (Direct Sequence Spread Spectrum) physical layer. 802.11a and 802.11g ratified respectively in 1999 and 2003, are based on OFDM (Orthogonal Frequency Division Multiplexing) physical layer, thereby allowing a peak physical rate of 54 Mbps. In 2009, the 802.11n called HT (High Throughput) WLAN is designed for achieving a peak physical

rate of 600 Mbps, by using 40 MHz channel bonding and MIMO (Multiple-Input Multiple-Output) spatial multiplexing. In 2013, 802.11ac known as VHT (Very High Throughput) WLAN has emerged for providing a peak physical rate up to 7 Gbps, by introducing 160 MHz channel bonding and Multi-User (MU) MIMO transmissions.

The evolution of material and several peripheral in different areas in recent years has led to excessive use of WLANs as to be saturated, since only one user and a single transmission are at the same time possible in the whole channel [5]. Thus, it is precisely in the purpose of efficiently deploying WLANs in high density areas that the new IEEE 802.11ax standard has been proposed. 802.11ax labeled HEW (High Efficiency WLAN) promises to allow a massive connection of users to WLAN, while reducing access delay and increasing individual throughput [6]. To meet these needs, several innovative techniques and methods have been incorporated at both PHY and MAC layer of the IEEE 802.11ax standard [7], namely: Full Duplex (FD) radio communications, Orthogonal Frequency Division Multiple Access (OFDMA), Down-Link (DL) and Up-Link (UL) MU MIMO transmissions, spatial reuse, etc. OFDMA is considered as the key feature of the 802.11ax since it effectively meets the expectations of users in high density WLANs and at a lower cost [8]. Indeed, OFDMA consists in splitting a radio channel into sub-channels, which are then grouped to form at most nine (09) Resources Units (RUs). The Access Point (AP) therefore affects these resources units to different stations in the WLAN for enabling independent and simultaneous transmissions between several wireless stations.

For optimizing the functioning of OFDMA technology, a numerous research works have been devoted in the available literature. The common goal of these studies is designing and implementing OFDMA based MAC protocols, through which the resources units are allocated and hence multiuser access is made possible. However, the basic principles and operating modes of these MAC protocols are completely different from each other. This brings into question the capacity of these MAC protocols to better use the OFDMA technology. For this reason, the goal of this paper is threefold: firstly, we investigate the existing OFDMA MAC protocols available in the literature in order to identify the benefits and drawbacks of each MAC protocol. Secondly, we propose a new classification of OFDMA MAC protocols on the basis of new criteria. Finally, we suggest some directions for future research.

The rest of this paper is organized as follows: we present the key features of the IEEE 802.11ax standard and give an overview of the OFDMA technology in Section II. We review the main existing OFDMA MAC protocols and propose a new classification of the latter in Section III. We conclude the paper and suggest new perspectives for future research in Section IV.

II. BACKGROUND

This section is devoted to describe the new IEEE 802.11ax standard. We give the main innovative features of the 802.11ax in Subsection II-A. Then, we present in depth the OFDMA technology in Subsection II-B.

A. Main Features of the IEEE 802.11ax Standard

In the following, we provide the major characteristics introduced in the new IEEE 802.11ax standard to allow the deployment of the next generation of high density WLANs [7]:

- **DL and UL OFDMA:** OFDMA is one of the most important features of the IEEE 802.11ax standard. In case of DL OFDMA, the AP transmits frames to several STAs (stations) during a single TXOP (transmission opportunity). This is possible by splitting a radio channel into several radio sub-channels, each frame is then transmitted independently over a single sub-channel. In the case of UL OFDMA, the same thing happens again, only this time the deferent STAs transmit simultaneously their frames to the AP.
- **DL and UL MU-MIMO:** DL MU-MIMO is originally proposed in the IEEE 802.11ac standard. an AP can transmit several frames to different stations over the whole channel by using the spatial multiplexing technique. The UL MU-MIMO is a new operating mode incorporated to the IEEE 802.11ax standard, which simultaneous transmissions from different stations to an AP. This is possible through the adoption of the space diversity technique.
- **1024-QAM (Quadrature Amplitude Modulation):** 1024-QAM is a highest MCS (Modulation and Coding Schema) technique, since it allows to encode 10 bits per OFDM symbol. Contrary to IEEE 802.11ac which can only encode 8 bits per OFDM symbol, because it uses a MCS technique based 256-QAM.
- **Resource Units (RU):** It is a new concept introduced for the first time in the WLANs based IEEE 802.11ax standard. A RU is formed by gathering a set of smaller subcarriers. It is allocated by an AP to a given wireless station for enabling independent transmission, i.e., without interference.
- **Target Wakeup Time (TWT):** It is a mechanism introduced in IEEE 802.11ax for reducing power consumption, knowing that it is originally proposed in the IEEE 802.11ah standard. TWT enhances the legacy PS (Power Save) mechanism existing in the previous IEEE 802.11 versions by allowing several stations the change to the sleep state.

- **Spatial Reuse:** It is a new concept proposed in the IEEE 802.11ax standard for overcoming the excessive use of WLANs. It consists to allocate the same radio channel or the same RU to multiple stations. This is possible by means of BBS (Basic Service Set) color technique.

B. Overview of the OFDMA Technology

The OFDM technique was introduced in legacy WLANs, it allows a single user transmission. However, the OFDMA technique proposed for the next generation WLAN enables multiuser transmissions. The key factor of OFDMA is that the spectrum uses several subcarriers, a set of subcarriers is grouped to form a resource unit (RU). The shorter RU in IEEE 802.11ax contains 26 subcarriers (or 2 MHz). Therefore, in a 20 MHz channel, there are at maximum 9 RUs of 26-subcarriers every one of them, they are noted RU-26 [7]. The technical characteristics of the OFDMA technology are summarized in Table I.

The OFDMA technology enables both uplink and downlink multiuser communications. The downlink multiuser communications already exist in the IEEE 802.11ac standard, but the uplink multiuser communications are only possible in IEEE 802.11ax. In downlink mode, the AP transmits during a TXOP multiple frames to different stations using different RUs, which are previously allocated to the stations in question. In uplink mode, different stations transmit simultaneously data over different RU to the AP. The AP is the station which managing the uplink and downlink communications and the allocation of RUs. It sends the beacon frame in the beginning of the transmission to inform each station about the following parameters: number of RUs, duration of frames transmission, number of spatial streams to be allocated to a given station.

TABLE I
TECHNICAL CHARACTERISTICS OF THE OFDMA TECHNOLOGY

Bounds	2,4 GHz and 5 GHz
Channels (MHz)	20, 40, 80, 80+80, 160
FFT sizes	256, 512, 1024, 2048
Subcarrier spacing	78.125 kHz
Symbol time	12.8 us
Cyclic prefix (us)	0.8, 1.6, 3.2
Highest Modulation	1024 QAM
Special stream	1-8

III. SURVEY ON OFDMA BASED MAC PROTOCOLS

In this section, we review the main existing MAC protocols designed for managing the OFDMA communications. At this time we take the opportunity to classify the different MAC protocols according to new criteria, namely: the channel access method (scheduled access or random access) and the form of OFDMA transmission (OFDM over bonding 20 MHz channels or OFDMA over splitting a 20 MHz channel). Consequently, we define four classes of OFDMA MAC protocols: (i) OFDMA scheduled access, (ii) OFDMA random access, (iii) OFDM scheduled access, and (iv) OFDM random access.

A. OFDMA Scheduled Access

In this subsection, we investigate the main existing MAC protocols based on an AP for allocating the radio resources provided at by the PHY layer, by means of OFDMA technology. These radio resources can be either simple subcarriers or a set of subcarriers called resources units, knowing that the subcarriers are obtained by dividing a 20 MHz channel into 78.125 KHz subchannels. The main works carried out include:

Jung and Lim in 2012 [9] have proposed an OFDMA MAC protocol called GC-OFDMA (Group Contention based OFDMA). The basic principle of the latter is illustrated in Figure 1, it is as follows: after a DIFS (Distributed Inter-Frame Space) duration, the AP sends a beacon frame for inviting the stations having an UL data stream to transmit a R-MSG frame over a randomly chosen subchannel. The stations that successfully transmit their R-MSG frames will then be grouped into Burst Groups (BGs). This is done through a specific frame called Group Allocation, sent by the AP. The stations in the same BG in turn send RTS frames to the AP for subchannels allocation. The AP responds by a Block CTS frame for designating the stations which can participate in the UL communication by assigning one or multiple subchannels. The proposed MAC protocol certainly allows to a large number of users to connect to the network, but the access delay can not be controlled since the transmission of R-MSG frame is done over a subchannel randomly chosen.

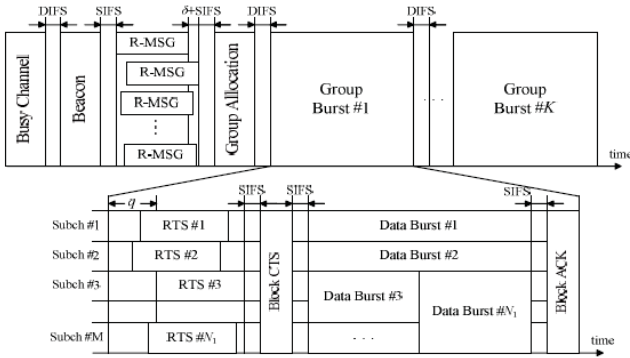


Fig. 1. GC-OFDMA MAC protocol [9]

Deng et al. in 2014 [10] designed an OFDMA MAC protocol said GRAP (Group Random Access Protocol) illustrated in Figure 2, it is based on PCF (Point Coordination Function) protocol and OFDMA channel access, in order to support diverse uses of wireless network. In GRAP, the AP senses the medium to be idle for a PIFS (PCF Inter-Frame Space) period, then broadcasts a beacon frame to inform the stations for the upcoming of the super-frame and the number of sub-channels. In the request period, the AP listens to all the random numbers at each subcarrier. In contention free period, the AP informs the stations by transmitting the CF-Poll (Contention Free Polling) frame to schedule the resource units for the stations in question. After a SIFS (Short Inter-Frame Space)

period, the stations transmit their packets simultaneously in uplink transmission. At the end, the AP broadcasts ACK frames and releases the channel. The proposed MAC protocol allows to better avoiding interferences on the chosen resources units. However, if a given resource unit is not selected by all the stations, it will not be used what constitute a waste of RUs.

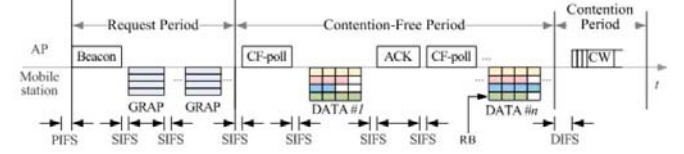


Fig. 2. GRAP protocol scheme [10]

Lee and Kim in 2017 [11] developed an HMAC (Hybrid Multiple Access Coordination) schema illustrated in Figure 3, it is based on AP for controlling the downlink and uplink transmissions. The AP senses the medium to be idle during an HIFS (Hybrid Inter-Frame Space) period: if it has data to transmit, it will then send a RMD (Request-to-Multiple-DL) frame over the entire channel. So, the AP informs the stations involved by the DL traffic, and at the same time indicates over which sub-channels the different transmissions will occur. The stations listed in the RMD frame, if they are ready for DL transmission, they use one sub-channel indicated in the RMD for transmitting their own CRU (Clear-to-Receive-with-UL-request) frames. The CRU frame is a preparation for DL transmission, and it can contain UL demands if it is needed. After a SIFS period, the AP sends data to stations in downlink mode, and receives an ACK after a SIFS. If the AP has no data to transmit, it sends null RMD without any station listed for CRU. For transmitting data in uplink mode, the AP sends a RMU (Request-to-Multiple-UL) frame to the stations that are previously sent the CRU frames during the control period. RMU frame allows to allocate sub-channels to the STAs and also schedules UL transmissions. The developed MAC protocol allows to minimize the overhead and eliminate the interferences. However, during the control period, the resource units which are allocated for a certain stations, can not be used by the other stations although the stations in question do not use them.

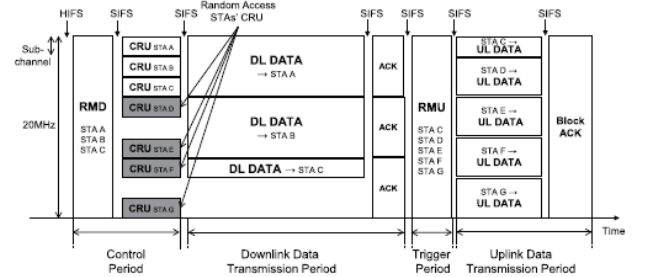


Fig. 3. HMAC protocol [11]

Lin *et al.* in 2018 [12] proposed a G-OFDMA (Group-based OFDMA) MAC protocol for multiuser access in WLAN. The protocol is illustrated in Figure 4, it operates as follows: the AP senses the medium to be idle for a DIFS period, it broadcasts a beacon frame. Then, in the CRR (Channel Resource Request) phase, each station randomly select one sub-channel and transmit its RTS frame during the RTS update interval. In GSA (Grouping and Sub-channel Allocation) phase, the AP groups the STAs, allocates sub-channels to the STAs that are successfully transmit their RTS frame in CRR phase, and broadcasts the BCTS (Block Clear-To-Send) frame to inform them. Finally, in DT (Data Transmission) phase, the STAs which are in the same group transmit their packets simultaneously. After a SIFS period, the AP broadcasts B-ACK. Like in GC-OFDMA protocol [9], the G-OFDMA protocol allows a massive connection of users. However, the fact that the AP does not affect a dedicate sub-channels for stations in a GB (Group Burst), this results in collisions during DT phase.

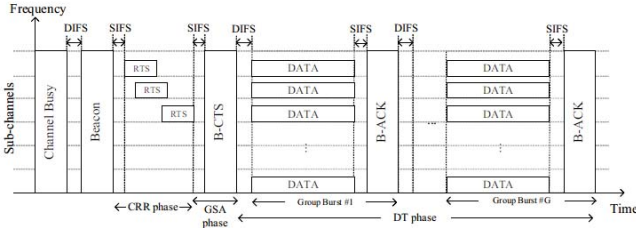


Fig. 4. G-OFDMA protocol [12]

B. OFDMA Random Access

In this subsection, we review the main existing OFDMA MAC protocols based on random access to the radio resources. These research works are:

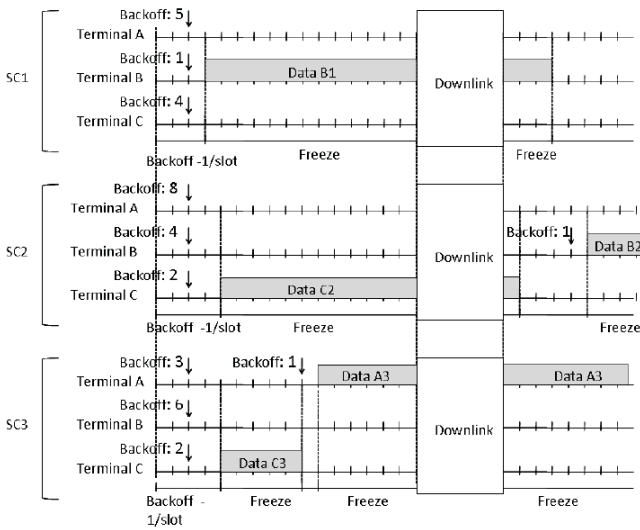


Fig. 5. An example of CM-CSMA/CA protocol [13]

Wang *et al.* in 2010 [13] proposed a CM-CSMA/CA (Concurrent Multichannel CSMA-CA: Carrier Sense Multiple Access Collision Avoidance) protocol illustrated in Figure 5 for with three stations and three subchannels. In this protocol, the number of subchannels is flexible to the number of terminals. The different stations maintain a backoff time counter for each subchannel, and decrement it by one slot time. The protocol also manages both uplink and downlink transmissions. The advantage of this protocol is that the interferences are avoided at maximum. However, the access procedure is very complicated since the stations should manage a backoff time counter for each subchannel.

Qu *et al.* in 2015 [14] proposed the OMAX (OFDMA based Multiple Access for IEEE 802.11ax) protocol for enabling simultaneous transmissions over various subchannels and hence enhancing the throughput. The OMAX is based on the DCF (Distributed Coordination Function) protocol and the enhanced RTS/CTS mechanism for reducing overheads. Figure 6 illustrates the proposed protocol: after sensing the whole channel to be idle for a DIFS period, the STAs execute a fast backoff; they decrement by number of idle sub-channels in each slot time. When the backoff around zero, the STAs in question randomly choose one subchannel, and send the RTS frame to the AP. At the end of request time, the AP broadcasts the G-CTS frame for allocating RUs to the STAs. After a SIFS period, the STAs cited in G-CTS frame transmit simultaneously, and then the AP replays with G-ACK frame. The concept of fast backoff introduced in OMAX protocol contribute to efficiently reducing the access delay to resources units. However, the loss of data persists since collisions can always occur.

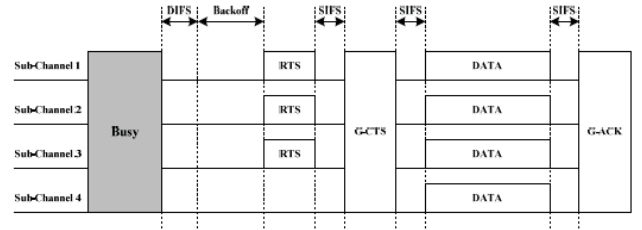


Fig. 6. OMAX protocol [14]

Lee in 2018 [15] proposed an OFDMA MAC protocol based random access illustrated in Figure 7. The AP sends the trigger frame request (TF-R) in the request period, and the STAs select randomly the RU in L stage. The AP transmits the trigger frame to the STAs and informs them about the allocated RU. Finally, the STAs transmit simultaneously their data on the allocated RU. In this protocol, the stations randomly choose the subchannels, thereby the RUs that are not selected, will not be used. This leads in an inefficient use in radio resources.

C. OFDM scheduled access

Shimamoto *et al.* in 2012 [16] proposed Two-Stage DCF-based access schema for OFDMA based WLANs. As illustrated in Figure 8, at the first DCF-based access stage,

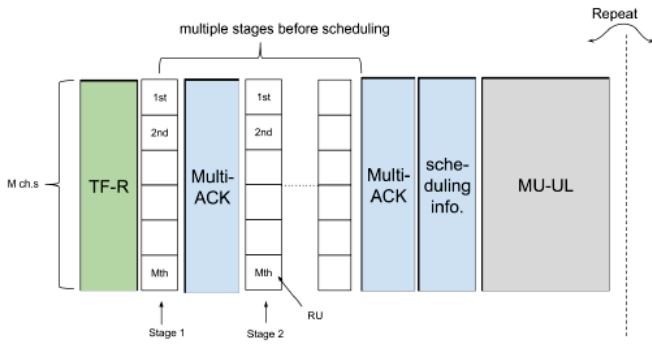


Fig. 7. Random access protocol for IEEE 802.11ax [15]

all the STAs that want to transmit the DATA, they execute the conventional DCF-based access. At the end of the timer backoff, the STA transmits the RTS frame to the AP, after that the AP broadcasts the CTS frame to the stations to inform them about the remaining subchannels. The AP uses the channel state information (CSI) contained in RTS frame to allocate the subchannels. In the second DCF-based access, the station contends for the remaining subchannels, then the AP broadcasts ARI (Assigned Resource Information) to schedule the subchannels for the stations and initiate the uplink transmissions. In this protocol, the number of STAs that can transmit simultaneously is just two. So, we suggest to add a number of DCF-based access stages equal to a number of subchannels, in order to efficiently use the radio channel.

D. OFDM random access

Islam *et al.* in 2018 [17] proposed the HTFA (High Throughput and Fair Access) protocol, where the operation rules are illustrated in Figure 9. HTFA consists in 2 steps: in the first step, the subchannels are evenly distributed to the stations. Then, in second step inside the sub-channel, the stations will contest for random access if the number of stations is larger than the number of sub-channels in the system. The second step depends of the number of STAs in the sub-channel and it uses the conventional DCF to select randomly the STA that will transmit. With this protocol, the spectrum is not used efficiently, because in step 1 the subchannel is reserved before random access in step 2. It will be benefit to inverse the two steps, i.e., selecting randomly the subchannel in the first step, and in second step grouping the STAs in deferent groups.

IV. CONCLUSIONS AND PROSPECTS

In this paper, we have focused our study on OFDMA technology introduced in the new IEEE 802.11ax standard for enabling the next generation of High density WLANs. OFDMA technology has the ability to allow a massive connection of users while reducing access delay and increasing individual throughput. So, to efficiently use the OFDMA technology, the challenge is designing at MAC layer protocols which meet the objectives of users in high density WLANs.

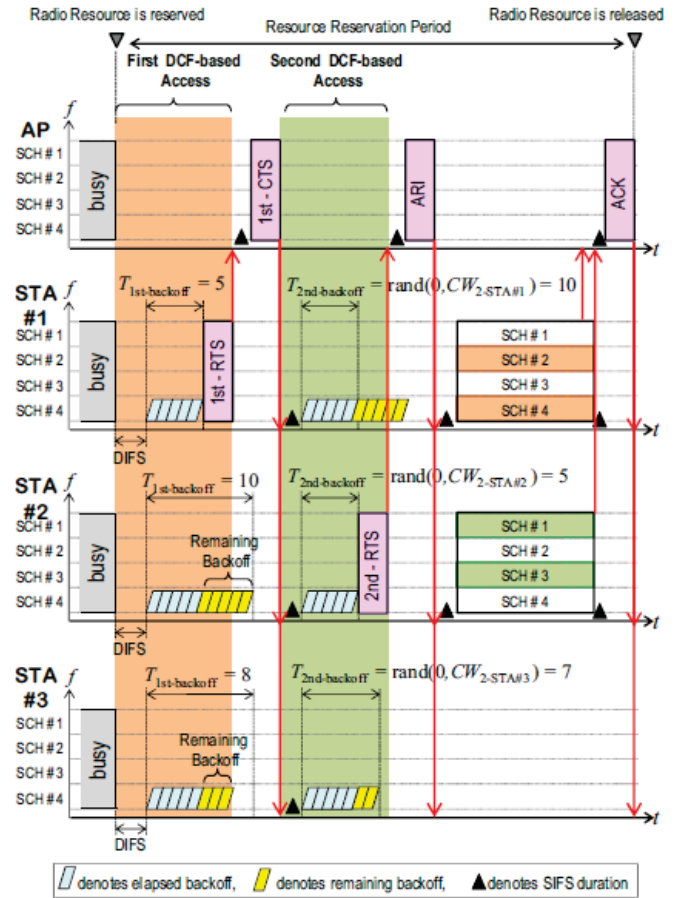


Fig. 8. Scheme based on Two-Stage DCF [16]

This is why, a numerous of research works have been devoted in the literature, which have as a common goal optimizing the use of radio resources provided at PHY layer by the OFDMA technology. In this paper, we have proposed a survey on MAC protocols based OFDMA technology with the aim of reviewing the main exiting MAC protocols, but also to elaborate a new classification of considered protocols. Through this study, we have been able to highlight the benefits and drawbacks of each studied protocol. In the following, we suggest the main recommendations to be considered for designing new OFDMA MAC protocols:

- The needs in term of uplink data streams of each station should be expressed in a deterministic manner.
- The radio resources should be used in an optimal manner.
- It should be taken into account of variable length of data.
- The access method to allocate the radio resources should be as possible simple.
- It will be interesting to design an hybrid MAC protocol based OFDMA in the sense of using a scheduled access for avoiding collisions and a random access for avoiding the waste of radio resources.

Sub-channel 1 Station B	
Sub-channel 2 Station A	
Sub-channel 3 Station C	
(a)	
Sub-channel 1 Station B	
Sub-channel 2 Station A	
Sub-channel 3 Station B	
(b)	
Sub-channel 1 Station B	Station C
Sub-channel 2 Station A	
Sub-channel 3 Station C	
(c)	

Fig. 9. Operation rules of HTFA protocol [17]

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